

Functional data analysis

7. Functional regression models

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Objective are:

- ▶ *to discuss the various forms of the functional linear model, including function on function regression or function/scalar regression;*
- ▶ *to present estimation methods and properties.*

Outline of the lecture

Introduction

The scalar response model

The functional response model

The fully functional model

Introduction

- ▶ The linear regression is perhaps the most useful and widely used statistical model.
- ▶ We can distinguish three cases, in which either the responses or the regressors, or both are curves.
- ▶ Implemented in the [refund](#) package.

► *The fully functional model:*

$$Y_i(t) = \mu(t) + \int_0^1 \psi(t, s) X_i(s) ds + \varepsilon_i(t), \quad t \in [0, 1].$$

In this model, the responses Y_i are curves, and so are the regressors X_i .

► *The scalar response model:*

$$Y_i = \alpha + \int_0^1 \beta(s) X_i(s) ds + \varepsilon_i, \quad t \in [0, 1].$$

In this model, the regressors are curves, but the responses are scalars.

► *The functional response model:*

$$Y_i(t) = \mu(t) + \rho(t) Z_i + \varepsilon_i(t), \quad t \in [0, 1].$$

In this model, the responses are curves, but the regressors are scalars.

The scalar response model

- ▶ Domains in which scalar-on-function regression has been applied include chemometrics, cardiology, brain science, climate science, and many others.
- ▶ We start with the model .

$$Y_i = \alpha + \int_0^1 \beta(s)X_i(s)ds + \varepsilon_i, \quad i = 1, \dots, n.$$

- ▶ Three strategies have been developed to deal with this model.
- ▶ The first two redefine the problem using a basis coefficient expansion of β .
- ▶ The third replaces the potentially high-dimensional covariate functions by a low-dimensional approximation using principal components.

Estimation through basis functions

- ▶ Let $(v_k(t))$ be a given set of basis functions.
- ▶ Assume that

$$\beta(t) = \sum_{k=1}^K c_k v_k(t).$$

- ▶ The number K of basis functions is generally chosen to be smaller than the number of time points t_j at which the functions X_i are observed. It reflects intuition as to how smooth the resulting estimate should be.
- ▶ There are data driven methods of selecting K .
- ▶ It is always useful to experiment with several selections of K .

- Using expansion of β , we obtain

$$\int_0^1 \beta(s) X_i(s) ds = \sum_{k=1}^K c_k \underbrace{\int_0^1 v_k(s) X_i(s) ds}_{x_{ik}} =: \sum_{k=1}^K x_{ik} c_k.$$

- The model reduces to

$$Y_i = \alpha + \sum_{k=1}^K x_{ik} c_k + \varepsilon_i, \quad i = 1, \dots, n.$$

- The parameter vector $\mathbf{c} = [\alpha, c_1, \dots, c_K]'$ is estimated by

$$\hat{\mathbf{c}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y},$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1K} \\ 1 & x_{21} & \dots & x_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{nK} \end{bmatrix}, \quad \mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}.$$

- ▶ A disadvantage of this method is that the resulting estimate

$$\hat{\beta}(s) = \sum_{k=1}^K \hat{c}_k v_k(s)$$

depends on the shape of the basis functions v_k and on their number K , both being subjectively chosen.

- ▶ The reduction to the standard model also leads to approximate confidence intervals for the unknown regression function β .
- ▶ If the errors, ε_i , are independent $\mathcal{N}(0, \sigma_\varepsilon^2)$, then the covariance matrix of $\hat{\mathbf{c}}$ is

$$\text{var}[\hat{\mathbf{c}}] = \sigma_\varepsilon^2 (\mathbf{X}'\mathbf{X})^{-1}.$$

- ▶ The error variance, σ_ε^2 , is estimated by the sample variance, $\hat{\sigma}_\varepsilon^2$, of the residuals

$$\hat{\varepsilon}_i = y_i - \hat{\alpha} - \sum_{k=1}^K x_{ik} \hat{c}_k.$$

- ▶ The variance of $\hat{c}_k$ is thus estimated by the corresponding diagonal entry of the matrix $\hat{\sigma}_\varepsilon^2(\mathbf{X}'\mathbf{X})^{-1}$.
- ▶ Denoting this estimate by $\hat{\sigma}_k^2$, an approximate 95% confidence interval for the regression function β is

$$\sum_{k=1}^K \hat{c}_k v_k(t) \pm 1.96 \sum_{k=1}^K \hat{\sigma}_k v_k(t).$$

- ▶ This confidence interval can be used only as an exploratory tool.

- ▶ The parameter K should be viewed as a tuning parameter.
- ▶ Adjusting K adjusts the smoothness of the resulting estimator $\hat{\beta}(t)$.
- ▶ From this perspective it is more desirable to smooth by using a roughness penalty approach.
- ▶ One estimates α and $\beta(t)$ by finding

$$(\hat{\alpha}, \hat{c}_k) := \underset{\alpha, (c_k)}{\operatorname{argmin}} \left\{ \sum_{i=1}^n \left(Y_i - \alpha - \sum_{k=1}^K x_{ik} c_k \right)^2 + \lambda P((c_k)) \right\},$$

for some $\lambda > 0$ and some penalty function $P(\cdot)$.

- ▶ Example of penalty: $P((c_k)) = \sum_{k=1}^K |c_k|$.

Regression on functional principal components

- ▶ Let

$$X_1(t), \dots, X_n(t) \stackrel{iid}{\sim} Sp(\mu, \gamma).$$

- ▶ Then

$$X_i(t) = \mu(t) + \sum_{j=1}^{\infty} \xi_{ij} v_j(t),$$

where (v_j) are the functional principal components, i.e. the eigenfunctions of the covariance operator γ .

- ▶ Denote by $\hat{v}_j$ the estimated functional principal components.
- ▶ In the FPC regression reduction to the standard model is achieved by using the approximation

$$X_i(t) \approx \hat{\mu}(t) + \sum_{j=1}^p \hat{\xi}_{ij} \hat{v}_j(t),$$

where

$$\hat{\xi}_{ij} = \int_0^1 [X_i(t) - \hat{\mu}(t)] \hat{v}_j(t) dt.$$

- Replacing the approximation by an equality, informally reduces the scalar response model to

$$\begin{aligned} Y_i &= \alpha + \int_0^1 \beta(t) \left(\hat{\mu}(t) + \sum_{j=1}^p \hat{\xi}_{ij} \hat{v}_j(t) \right) dt + \varepsilon_j \\ &= \beta_0 + \sum_{j=1}^p \hat{\xi}_{ij} \beta_j + \varepsilon_i, \end{aligned}$$

where

$$\beta_0 = \alpha + \int_0^1 \beta(t) \hat{\mu}(t) dt, \quad \beta_j = \int_0^1 \beta(t) \hat{v}_j(t) dt$$

are treated as unknown parameters.

- Define

$$\mathbf{Z} = \begin{bmatrix} 1 & \hat{\xi}_{11} & \dots & \hat{\xi}_{1K} \\ 1 & \hat{\xi}_{21} & \dots & \hat{\xi}_{2K} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \hat{\xi}_{n1} & \dots & \hat{\xi}_{nK} \end{bmatrix},$$

estimate the parameter $\beta = [\beta_0, \beta_1, \dots, \beta_p]'$ from the regression

$$\mathbf{Y} = \mathbf{Z}\beta + \mathbf{e}.$$

- Hence,

$$\hat{\beta} = [\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p]' = (\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'\mathbf{Y}$$

and

$$\hat{\beta}(t) = \sum_{j=1}^p \hat{\beta}_j \hat{v}_j(t), \quad \hat{\alpha} = \hat{\beta}_0 - \sum_{j=1}^p \hat{\beta}_j \int_0^1 \hat{v}_j(t) \hat{\mu}_j(t) dt.$$

Functional polynomial regression

- ▶ These are linear in the coefficients but not in the predictors, in contrast to the models, which are not linear in either sense.
- ▶ The functional quadratic regression model can be expressed as

$$Y_i = \alpha + \int_0^1 X_i(t)\beta(t)dt + \int_0^1 \int_0^1 X_i(s)X_i(t)\delta(s, t)dsdt + \varepsilon_i.$$

- ▶ Here we have both linear and quadratic coefficient functions, $\beta(t)$ and $\delta(s, t)$.

Nonlinear scalar-on-function regression

- ▶ In many applications, the assumption of a linear relationship between X and Y is too restrictive to describe the data.
- ▶ Single-index model:

$$Y_i = h\left(\int_0^1 X_i(t)\beta(t)dt\right) + \varepsilon_i, \quad i = 1, \dots, n$$

allowing the function $h(\cdot)$ to be any smooth function defined on the real line.

- ▶ Multiple-index model

$$Y_i = \sum_{j=1}^J h_j\left(\int_0^1 X_i(t)\beta_j(t)dt\right) + \varepsilon_i.$$

Nonparametric scalar response model

- ▶ Nonparametric model has a representation

$$Y_i = m(X_i) + \varepsilon_i,$$

where $m : L_2(0, 1) \rightarrow \mathbb{R}$ is a functional that must be estimated.

- ▶ Approaches to estimating m extend traditional non-parametric regression methods.
- ▶ Functional Nadaraya-Watson estimator is

$$\hat{m}(x) = \frac{\sum_{i=1}^n K[d(x, X_i)/h] Y_i}{\sum_{i=1}^n K[d(x, X_i)/h]},$$

where $K(\cdot)$ is a kernel function, which we define as a function supported and decreasing on $[0, 1)$; $h > 0$ is a bandwidth, and $d(x, y)$ is a semi-metric.

- The performance of the functional NW estimator can depend crucially on the chosen semi-metric d .
- For smooth functional data, it may be appropriate to use the derivative-based semi-metric

$$d_q(x, y) = \left(\int_0^1 [x^{(q)}(t) - y^{(q)}(t)]^2 dt \right)^{1/2},$$

where $x^{(q)}$ being the q 'th-order derivative of x .

- In practice $q = 2$ is a popular choice.

The functional response model

- The function-on-scalar regression model is

$$Y_i(t) = \beta_1(t)Z_{1i} + \beta_2(t)Z_{2i} + \cdots + \beta_q(t)Z_{qi} + \varepsilon_i(t), \quad i = 1, \dots, n$$

where (Z_{ki}) are scalar regressors and β_k are unknown parameters.

- The q functional regression parameters β_k are often called the *effect functions*.

► Setting

$$\mathbf{Y}(t) = \begin{bmatrix} Y_1(t) \\ Y_2(t) \\ \vdots \\ Y_n(t) \end{bmatrix}, \quad \mathbf{Z} = \begin{bmatrix} Z_{11} & Z_{21} & \dots & Z_{q1} \\ Z_{12} & Z_{22} & \dots & Z_{q2} \\ \vdots & \vdots & \ddots & \vdots \\ Z_{1n} & Z_{2n} & \dots & Z_{qn} \end{bmatrix}$$

and

$$\boldsymbol{\beta}(t) = \begin{bmatrix} \beta_1(t) \\ \beta_2(t) \\ \vdots \\ \beta_q(t) \end{bmatrix}, \quad \boldsymbol{\varepsilon}(t) = \begin{bmatrix} \varepsilon_1(t) \\ \varepsilon_2(t) \\ \vdots \\ \varepsilon_n(t) \end{bmatrix}$$

the model can be written as

$$\mathbf{Y}(t) = \mathbf{Z}\boldsymbol{\beta}(t) + \boldsymbol{\varepsilon}(t).$$

- The least square estimator of the parameter β is defined as

$$\begin{aligned}\hat{\beta} &= \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^n \left\| Y_i - \sum_{k=1}^q Z_{ki} \beta_k \right\|^2 \right\} \\ &= \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^n \int_0^1 e_i^2(\beta, t) dt \right\},\end{aligned}$$

- where

$$e_i(\beta, t) = Y_i(t) - \sum_{k=1}^q Z_{ki} \beta_k(t).$$

- For each fixed t the function $\sum_{i=1}^n e_i^2(\beta, t)$ is minimized by

$$\hat{\beta}(t) = (\mathbf{Z}'\mathbf{Z})^{-1} \mathbf{Z}'\mathbf{Y}(t).$$

- This is also the solution of minimization problem.

- The fitted (predicted) values of response and residuals are computed as

$$\hat{\mathbf{Y}}(t) = \mathbf{Z}\hat{\boldsymbol{\beta}}(t), \quad \hat{\boldsymbol{\varepsilon}}(t) = \mathbf{Y}(t) - \hat{\mathbf{Y}}(t).$$

The fully functional model

Recall:

$$Y_i(t) = \int \beta(t, s) X_i(s) ds + \varepsilon_i(t), \quad i = 1, \dots, n.$$

- ▶ The value $\beta(t, s)$ reflects the effect of the explanatory function X_i at time s on the response function Y_i at time t .
- ▶ To develop an estimation procedure analogous to the least squares estimation in classical model, and implemented in the R package `fda`, the model is rewritten in the following form:

$$\mathbf{Y}(t) = \int \mathbf{X}(s) \beta(s, t) ds + \boldsymbol{\varepsilon}(t),$$

where where

$$\begin{aligned}\mathbf{Y}(t) &= [Y_1(t), Y_2(t), \dots, Y_n(t)]', \\ \mathbf{X}(s) &= [X_1(s), X_2(s), \dots, X_n(s)]', \\ \boldsymbol{\varepsilon}(t) &= [\varepsilon_1(t), \varepsilon_2(t), \dots, \varepsilon_n(t)]'.\end{aligned}$$

- ▶ Suppose $\{\eta_k, k \geq 1\}$ and $\{\theta_k, k \geq 1\}$ are some bases, for example Fourier and spline, which need not be orthonormal.
- ▶ The functions η_k are suitable for expanding the functions X_i and the θ_i for expanding the Y_i .
- ▶ The idea of the estimation of the kernel $\beta(\cdot, \cdot)$ is to consider estimates of the form

$$\beta^*(s, t) = \sum_{k=1}^K \sum_{\ell=1}^L b_{k\ell} \eta_k(s) \theta_\ell(t)$$

in which K and L are relatively small numbers which are used as smoothing parameters; the smaller K and L , the smoother the estimate of $\beta(\cdot, \cdot)$.

- ▶ A least squares estimator is then obtained by finding $b_{k\ell}$ which minimize the residual sum of squares:

$$\sum_{i=1}^n \|Y_i - \int X_i(s)\beta^*(s, \cdot)\|^2.$$

- ▶ Consistency properties of this approach are not fully understood, but it gives useful estimates, which can be computed analogously to the standard vector case.

- ▶ Using EFPC's rather than fixed bases offers another approach to the estimation of $\psi(\cdot, \cdot)$.
- ▶ Consider two L_2 -valued mean zero random functions X and Y , and their expansions

$$X(s) = \sum_{i=1}^{\infty} \xi_i v_i(s), \quad Y(t) = \sum_{j=1}^{\infty} \zeta_j u_j(t),$$

where the v_j are the FPC's of X and the u_j the FPC's of Y and

$$\xi_i = \langle X, v_i \rangle, \quad \zeta_j = \langle Y, u_j \rangle.$$

Lemma

Suppose $X, Y, \varepsilon \in L_2$ are mean zero, ε is independent of X , and the following linear equation hold

$$Y(t) = \int \psi(t, s)X(s)ds + \varepsilon(t),$$

with the kernel $\psi(\cdot, \cdot)$ satisfying

$$\iint \psi^2(t, s)dt ds < \infty.$$

Then

$$\psi(t, s) = \sum_{k=1}^{\infty} \sum_{\ell=1}^{\infty} \frac{E[\xi_{\ell} \zeta_k]}{E[\xi_{\ell}^2]} u_k(t) v_{\ell}(s),$$

where the convergence is in $L_2([0, 1] \times [0, 1])$.

Remark

Observe that $E[\xi_\ell^2] = \lambda_\ell$ is the eigenvalue corresponding to v_ℓ . The eigenfunctions v_ℓ belonging to zero eigenvalues can be omitted from representation without changing it (it is an L_2 representation), so we may assume that $E[\xi_\ell^2] > 0$ for each $\ell \geq 1$.

- Lemma suggest the following estimator:

$$\hat{\psi}_{KL}(t, s) = \sum_{k=1}^K \sum_{\ell=1}^L \hat{\lambda}_{\ell}^{-1} \hat{\sigma}_{\ell k} \hat{u}_k(t) \hat{v}_{\ell}(s),$$

where $\hat{\sigma}_{\ell k}$ is an estimator of $E[\xi_{\ell} \zeta_k]$.

- The simplest estimator is

$$\hat{\sigma}_{\ell k} = n^{-1} \sum_{i=1}^n \langle X_i, \hat{v}_{\ell} \rangle \langle Y_i, \hat{u}_k \rangle.$$